

Name: Xavier Nathaniel Umbaligan

Date: 8/11/2026

Section: IT703P

Instructor: Jonell Franzlvero

Activity 5: Computer Startup Problems Research

Objective:

Identify common problems that may occur during the computer boot process and determine their possible causes and solutions.

Instructions:

Research online for common problems encountered when turning on a computer. Find at least 5 different problems. Then create a table and must provide and answer the following:

1. Problem Encountered
2. Possible Cause
3. Boot Process Stage Involved
4. Possible Solution

Problem Encountered	Possible Cause	Boot Process Stage Involved	Possible Solution
Computer does not power on	Faulty power supply, loose power cable, dead battery	Power-On Stage	Check power connections, test the power supply, or replace the battery.
Black screen / No display	Monitor not connected, faulty graphics card, display cable issue	POST (Power-On Self-Test)	Check monitor connections, reseal the graphics card, or replace the cable.
Continuous beeping during startup	Defective RAM, motherboard, or hardware component	POST (Power-On Self-Test)	Reseat or replace the RAM and check hardware connections.

"Operating System Not Found" error	Incorrect boot order, disconnected drive, corrupted boot records	Boot Loader Stage	Verify boot order in BIOS/UEFI, reconnect the drive, or repair boot files.
Windows stuck on loading screen	Corrupted system files, faulty drivers, failed updates	OS Loader / Kernel Stage	Run Startup Repair, boot into Safe Mode, or restore the system.

Guide Questions:

After completing the table, answer:

1. Which startup problem did you find most difficult to diagnose? Why?

The black screen/no display problem was the most difficult to diagnose because it can be caused by several issues, such as a faulty monitor, loose cables, graphics card problems, or motherboard failure.

2. Which boot process stage is associated with the most problems you found?

The POST (Power-On Self-Test) stage is associated with the most problems because many hardware-related issues, such as RAM, motherboard, and display errors, occur during this stage.

3. What problem would prevent the computer from reaching the operating system?

The "Operating System Not Found" error would prevent the computer from reaching the operating system because the system cannot locate the files needed to start the OS.

4. What problem would occur after the operating system has already started loading?

A Windows stuck on the loading screen problem would occur after the operating system has already started loading, usually due to corrupted system files or faulty drivers.

5. Why is understanding the boot process important when troubleshooting a computer?

Understanding the boot process is important because it helps identify where a problem occurs, making it easier to diagnose and fix startup issues efficiently.